

Prefab Definitions

This document describes all prefabs needed for the Space Shooter game.

Player Prefabs

Player Ship

File: Assets/Prefabs/Player.prefab

```

GameObject: Player
  Tag: "Player"
  Layer: "Player"
  Components:
    Transform
    Position: (0, -3, 0)
    SpriteRenderer
    Sprite: player_ship.png
    Color: White
    Sorting Layer: "Characters" (Order: 10)
    Rigidbody2D
    Body Type: Dynamic
    Gravity Scale: 0
    Freeze Rotation: Z = true
    BoxCollider2D or PolygonCollider2D
    Is Trigger: true
    PlayerController (Script)
    Move Speed: 8
    Fire Rate: 0.2
    Bullet Prefab: PlayerBullet
    Fire Point: FirePoint child object
    HealthSystem (Script)
    Max Health: 100
    AudioSource
    Play On Awake: false
  Children:
    FirePoint
    Position: (0, 0.5, 0)

```

Player Bullet

File: Assets/Prefabs/PlayerBullet.prefab

```

GameObject: PlayerBullet
Tag: "PlayerBullet"
Layer: "PlayerBullets"
Components:
  Transform
  SpriteRenderer
    Sprite: bullet_player.png
    Color: Cyan (0, 1, 1, 1)
    Sorting Layer: "Projectiles" (Order: 5)
  Rigidbody2D
    Body Type: Dynamic
    Gravity Scale: 0
    Collision Detection: Continuous
  CapsuleCollider2D or CircleCollider2D
    Is Trigger: true
  Bullet (Script)
    Speed: 12
    Damage: 1
    Is Player Bullet: true
    Lifetime: 5

```

Enemy Prefabs

Basic Enemy

File: Assets/Prefabs/BasicEnemy.prefab

```

GameObject: BasicEnemy
Tag: "Enemy"
Layer: "Enemies"
Components:
  SpriteRenderer
    Sprite: enemy_basic.png
    Color: Red (1, 0.3, 0.3, 1)
    Sorting Layer: "Characters" (Order: 5)
  Rigidbody2D
    Body Type: Kinematic
    Gravity Scale: 0
  BoxCollider2D or CircleCollider2D
    Is Trigger: true
  EnemyBase (Script)
    Enemy Type: Basic
    Move Speed: 3
    Score Value: 100
    Can Shoot: true
    Fire Rate: 2
    Bullet Prefab: EnemyBullet
  HealthSystem (Script)
    Max Health: 20

```

Zigzag Enemy

File: Assets/Prefabs/ZigzagEnemy.prefab

GameObject: ZigzagEnemy

```

[ ] Tag: "Enemy"
[ ] Components:
[ ]   [ ] SpriteRenderer
[ ]     [ ] Color: Orange (1, 0.6, 0.2, 1)
[ ]     [ ] EnemyBase (Script)
[ ]       [ ] Enemy Type: Zigzag
[ ]       [ ] Move Speed: 4
[ ]       [ ] Score Value: 150
[ ]       [ ] Zigzag Amplitude: 2
[ ]       [ ] Zigzag Frequency: 2
[ ]     [ ] HealthSystem
[ ]       [ ] Max Health: 15

```

Circular Enemy

File: Assets/Prefabs/CircularEnemy.prefab

GameObject: CircularEnemy

```

[ ] Tag: "Enemy"
[ ] Components:
[ ]   [ ] SpriteRenderer
[ ]     [ ] Color: Purple (0.8, 0.3, 1, 1)
[ ]     [ ] EnemyBase (Script)
[ ]       [ ] Enemy Type: Circular
[ ]       [ ] Move Speed: 2
[ ]       [ ] Score Value: 200
[ ]       [ ] Circular Radius: 1.5
[ ]       [ ] Circular Speed: 3
[ ]     [ ] HealthSystem
[ ]       [ ] Max Health: 25

```

Charger Enemy

File: Assets/Prefabs/ChargerEnemy.prefab

GameObject: ChargerEnemy

```

[ ] Tag: "Enemy"
[ ] Components:
[ ]   [ ] SpriteRenderer
[ ]     [ ] Color: Yellow (1, 1, 0.3, 1)
[ ]     [ ] EnemyBase (Script)
[ ]       [ ] Enemy Type: Charger
[ ]       [ ] Move Speed: 5
[ ]       [ ] Score Value: 250
[ ]       [ ] Can Shoot: false
[ ]     [ ] HealthSystem
[ ]       [ ] Max Health: 10

```

Boss Enemy

File: Assets/Prefabs/BossEnemy.prefab

```

GameObject: BossEnemy
  Tag: "Enemy"
  Components:
    SpriteRenderer
    Sprite: enemy_boss.png (larger sprite)
    Color: Dark Red (0.8, 0, 0, 1)
    BossEnemy (Script)
    Move Speed: 1.5
    Score Value: 1000
    Fire Rate: 1
    Spread Shot Count: 5
    Circular Shot Count: 12
    HealthSystem
    Max Health: 500

```

Enemy Bullet

File: Assets/Prefabs/EnemyBullet.prefab

```

GameObject: EnemyBullet
  Tag: "EnemyBullet"
  Layer: "EnemyBullets"
  Components:
    SpriteRenderer
    Color: Red (1, 0.3, 0.3, 1)
    Sorting Layer: "Projectiles"
    Rigidbody2D
    Gravity Scale: 0
    CircleCollider2D
    Is Trigger: true
    Bullet (Script)
    Speed: 6
    Damage: 1
    Is Player Bullet: false
    Lifetime: 5

```

Power-Up Prefabs

PowerUp

File: Assets/Prefabs/PowerUp.prefab

```

GameObject: PowerUp
  Tag: "PowerUp"
  Layer: "PowerUps"
  Components:
    SpriteRenderer
    Sprite: powerup.png (diamond or star shape)
    Sorting Layer: "Pickups" (Order: 8)
    CircleCollider2D
    Is Trigger: true
    Radius: 0.3
    PowerUp (Script)
    Move Speed: 2
    Lifetime: 10
    Duration: 5 (for timed power-ups)
    Amount: 25 (for health/score)

```

Manager Prefabs

GameManager

File: Assets/Prefabs/GameManager.prefab

```

GameObject: GameManager
  Components:
    GameManager (Script)
    Starting Lives: 3
    Invincibility Duration: 2
    ScoreManager (Script)
    AudioManager (Script)
    Music Source: (child AudioSource)
    SFX Source: (child AudioSource)
  Children:
    MusicSource (AudioSource component)
    SFXSource (AudioSource component)

```

ObjectPooler

File: Assets/Prefabs/ObjectPooler.prefab

```

GameObject: ObjectPooler
  Components:
    ObjectPooler (Script)
    Pools:
      Tag: "PlayerBullet", Size: 30
      Tag: "EnemyBullet", Size: 50
      Tag: "BasicEnemy", Size: 20
      Tag: "ZigzagEnemy", Size: 15
      Tag: "CircularEnemy", Size: 15
      Tag: "ChargerEnemy", Size: 10
      Tag: "PowerUp", Size: 10

```

Background Prefabs

ParallaxBackground

File: Assets/Prefabs/ParallaxBackground.prefab

```
GameObject: ParallaxBackground
├── Components:
│   ├── ParallaxBackground (Script)
│   │   └── Scroll Speed: 0.5
│   ├── StarfieldGenerator (Script)
│   │   ├── Star Count: 100
│   │   ├── Field Width: 20
│   │   └── Field Height: 15
│   └── Children:
│       ├── BackgroundLayer1 (furthest, slowest)
│       │   └── SpriteRenderer (space background)
│       ├── BackgroundLayer2 (closer, faster)
│       │   └── SpriteRenderer (nebula/stars overlay)
```

UI Prefabs

Canvas

The Canvas should be set up in each scene with the UIManager component:

```
Canvas
├── Canvas Scaler
│   ├── UI Scale Mode: Scale With Screen Size
│   ├── Reference Resolution: 1920 x 1080
│   └── Match: 0.5
├── UIManager (Script)
└── Children:
    ├── HUDPanel
    │   ├── ScoreText (TextMeshPro)
    │   ├── HighScoreText (TextMeshPro)
    │   ├── WaveText (TextMeshPro)
    │   ├── LivesText (TextMeshPro)
    │   ├── HealthBar (Slider)
    │   ├── MultiplierText (TextMeshPro)
    │   └── ComboText (TextMeshPro)
    ├── PausePanel
    │   ├── Title: "PAUSED"
    │   ├── ResumeButton
    │   ├── RestartButton
    │   └── MainMenuButton
    ├── GameOverPanel
    │   ├── Title: "GAME OVER"
    │   ├── FinalScoreText
    │   ├── FinalWaveText
    │   ├── HighScoreText
    │   ├── RestartButton
    │   └── MainMenuButton
    └── VictoryPanel
        ├── Title: "VICTORY!"
        ├── FinalScoreText
        ├── RestartButton
        └── MainMenuButton
```